

Web services - Retrofit

Contents

- **Web services**
- **Retrofit**

Android web services



Retrofit

- A type-safe REST client for Android developed by Square
- Easy to retrieve and upload JSON (or other structured data) via a REST based webservice
- Set up

```
dependencies {  
  
    implementation 'com.squareup.retrofit2:retrofit:2.0.2'  
    implementation 'com.squareup.retrofit2:converter-gson:2.0.2'  
}
```

HTTP Methods

- POST creates a resource on the server .
- GET retrieves an appropriate resource .
- PUT should be used to update the resource or change the state .
- DELETE is used only to remove the resource .

Method Annotations

- `@Body`: sends Java objects as request body.
- `@Path`: variable substitution for the API endpoint
- `@Url`: use dynamic URLs.
- `@Query`: To URL encode a query use the form
- `@Field`: send data as form-urlencoded. The `@Field` parameter works only with a POST

Retrofit Implementation steps

- Model class which is used as a JSON model
- Interfaces that define the possible HTTP operations
- Retrofit.Builder class - Instance which uses the interface and the Builder API to allow defining the URL end point for the HTTP operations.

Retrofit Implementation

- Creating the API Interface

```
public interface MyApiEndpointInterface {  
  
    @GET("users/{username}")  
    Call<User> getUser(@Path("username") String username);  
  
    @GET("group/{id}/users")  
    Call<List<User>> groupList(@Path("id") int groupId, @Query("sort") String sort);  
  
    @POST("users/new")  
    Call<User> createUser(@Body User user);  
  
}
```

Retrofit Implementation

- Create Java Classes for Resources
Using jsonschema2pojo
- Creating the Retrofit instance

```
public static final String BASE_URL = "http://api.myservice.com/";
```

```
Retrofit retrofit = new Retrofit.Builder().baseUrl(BASE_URL)
```

```
.addConverterFactory(GsonConverterFactory.create()).build();
```

```
MyApiEndpointInterface myRetrofitAPI = retrofit.create(MyApiEndpointInterface.class);
```

Retrofit Implementation

- Call

```
myRetrofitAPI.getUsers("12").enqueue(new Callback<User>() {  
    @Override  
    public void onResponse(Call<User> call, Response<User> response) {  
        // Success  
    }  
    @Override  
    public void onFailure(Call<User> call, Throwable t) {  
        //Failed  
    }  
});
```

Classwork

- Create a List screen using RecyclerView
- Load data from api: <https://jsonplaceholder.typicode.com/posts>
- Show data on RecyclerView

(More apis: <https://jsonplaceholder.typicode.com/>)

References

- <https://guides.codepath.com/android/consuming-apis-with-retrofit>
- <https://jsonplaceholder.typicode.com/>